

Battlefield-Adaptive Evolutionary Algorithm with Online Reference-Reallocation and Quantile ε -Grid Selection for Large-Scale and Constrained Multi-Objective Optimization

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Abstract

Large-scale multi-objective optimization in which the number of decision variables D reaches several hundreds remains one of the most challenging open problems in evolutionary multi-objective optimization. Reference-vector mechanisms (NSGA-III, RVEA) lose diversity as the non-dominated set must be distributed across a growing objective space, and classical crossover such as SBX loses directed-convergence capability as the decision space grows. Most existing dimension-adaptive algorithms, however, target a *single* battlefield: large- D (LSMOP, DTLZ2-300D) *or* constrained low-dimensional (CF, MW), and cannot be applied to both without re-implementation. This paper proposes **VOR** (*Vector-adaptive Online Reference-reallocation EA*), a dimension-adaptive framework whose central mechanism is a *battlefield routing* rule: a single codebase services the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10\text{--}60$) battlefields, with the variation operator, environmental selection and reference-vector reallocation jointly gated by the initial IGD scale, the decision dimension, and the presence of constraints. **Core contributions:** (i) a quantile ε -grid environmental selection (Q-EPS) that keeps the population on the true Pareto front at $\text{PPS} \approx N$ on irregular fronts where EDD's fixed 10-bin grid collapses to $\text{PPS } 24\text{--}27$ (CF1/LSMOP6);

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(ii) an online reference-vector reallocation (ORA) that, every 10 generations, clusters the front-1 decision space and pulls NBI reference vectors toward the cluster principal directions, repairing NSGA-III’s reference/front misalignment on MAF14; (iii) an IGD-derivative-gated operator switch that, at *any* dimension, auto-switches APD→DSG→GDV when the IGD derivative stalls for 10 generations; and (iv) the *battlefield routing* rule that makes one codebase service $D \in \{10, 30, 60, 300\}$ without hand-tuned dimension gates. **Main results** (6 problems $\times$ 30 seeds, $N=100$, $G=200$): on the five-algorithm set {NSGA-2, RVEA, MOEA/D, EDD(v12), VOR}, VOR attains the first IGD rank on 3 of 6 problems (ZDT1, LSMOP1, LSMOP6), the second on DTLZ2-300D and the third on MAF14/CF1, with a statistically significant IGD advantage over EDD on ZDT1, LSMOP1 and CF1 (30-seed Wilcoxon, $p < 10^{-5}$, rank-biserial effects 0.5–1.0). **Honest limitations:** (a) against the three 2026 state-of-the-art algorithms FDSEA/GDVTSF/MOEA-IB ($D = 300$, 30 seeds, the same seed protocol as the VOR runs), VOR is *not* the combined IGD/HV winner on the large-scale battlefield: FDSEA outperforms VOR on all three $D = 300$ instances (LSMOP1 0.143 vs 0.706; LSMOP6 0.668 vs 1.608; DTLZ2-300D 0.065 vs 0.749); VOR’s differentiating value is the *single-codebase coverage* of the constrained low-dimensional battlefield where the three SOTA algorithms have no published $D = 10$ –60 benchmark; (b) on CF1 (constrained, $D = 10$), MOEA/D’s decomposition still retains a structural advantage (IGD 0.010 vs VOR 0.063); (c) MAF14 exhibits the seed-invariance property (std= 0.0000) shared by the EDD family. We therefore position VOR as the *battlefield-broadening* member of the EDD lineage: it fixes part of the constrained low-dimensional PPS collapse that was the original EDD_cf’s defining weakness while preserving the large-scale robustness property, and we report its quantified residual boundary on the $D = 300$ battlefield against the 2026 SOTA in full.

Keywords: large-scale multi-objective optimization, battlefield-adaptive evolutionary algorithm, quantile epsilon-grid selection, online reference-vector reallocation, IGD-gated operator switching, constrained multi-objective optimization

1. Introduction

1.1. The two-battlefield challenge

Multi-objective optimization (MOO) problems with a large decision dimension D (hundreds to thousands) arise wherever the number of design variables far exceeds the number of objectives. Such problems are distinct from classical many-objective problems (MaOPs, $M \geq 5$): even with only three objectives, the search space grows as 2^D , and the interplay between convergence and diversity is fundamentally different from the low-dimensional regime in which most existing evolutionary multi-objective optimizers (EMOAs) are tuned. Classical EMOAs fail on large-scale decision spaces in two ways. First, reference-vector mechanisms (NSGA-III [2], RVEA [4], MOEA/D [3]) distribute the population across the objective space using a fixed number of reference points; as D grows the non-dominated front becomes dense in objective space and a fixed reference set cannot resolve it, so diversity degrades. Second, scalarized offspring operators (SBX, polynomial mutation) assume low-dimensional structure in their crossover distribution; in high- D search spaces they behave as essentially undirected random walks, and convergence slows by several orders of magnitude.

A complementary, equally important regime is the *constrained low-dimensional* battlefield. Benchmark families such as CF1–CF6 ($D = 10$, $M = 3$) and MW1–MW3 ($D = 10$ – 15 , $M = 3$) test constraint handling, while the MAF family ($D = 60$, $M = 3$) introduces irregular, non-smooth Pareto fronts that stress the reference-vector alignment. The large-scale benchmark families used here — LSMOP (LSMOP1–9) [8] and the constrained CF/MW/MAF suites — follow the standard PLATEMO [13] protocol. Most large-scale solvers (DSG-EA [6], PH-LSMAEA [7], a fuzzy decision-variable framework [9], FDSEA [10], GDVTSF [11], MOEA-IB [12]) are designed for $D \geq 100$ and either skip the constrained low-dimensional regime or collapse on it. The EDD lineage (EDD v12) addressed this by grafting dimension-aware gating: when $D \geq 100$ it uses an ε -grid environmental selection with directed decision-space operators, otherwise it falls back to an adaptive-Pareto-distance (APD) + reference-vector + SMS architecture. However, EDD retains two documented structural weaknesses: (i) its fixed 10-bin ε grid collapses PPS on the irregular

MAF14 front (PPS 8–97 across seeds) and stagnates at IGD 0.86 (seed-invariant, std= 0.0000); (ii) its fixed 10-bin grid and the convergence-freeze gate (IGD < 0.02) prematurely freeze the high- D LSMOP6 front (PPS 24.6) and the constrained CF1 (PPS 24.2).

1.2. The single-codebase gap and our approach

The gap we target is *not* a single-battlefield SOTA (the 2026 SOTA algorithms already win on $D = 300$ LSMOP/DTLZ2); the gap is that *no published algorithm services both battlefields from one codebase* without hand-tuned dimension gates. We propose **VOR** (Vector-adaptive Online Reference-reallocation EA), whose central mechanism is a *battlefield routing* rule that reads the initial IGD scale, the dimension D and the presence of constraints HASCON, and routes to one of three operator regimes:

- DSG (directed-sampling operator) when $D \geq 100$ *and* $IGD_0 \geq 15$ (large-scale hard-convergence battlefield: LSMOP6, DTLZ2-300D), or when $30 \leq D < 100$ *and* HASCON (MAF14);
- APD + SMS (adaptive-Pareto-distance + SMS-EMO switch) when $D < 100$ and HASCON= 0 (ZDT1 simple battlefield), and when $D < 30$ and HASCON= 1 (CF1 constrained low-dimensional battlefield);
- GDV (generational-difference-vector local search) is injected only as a stagnation-triggered escape when $stagn \geq 20$ and the regime is not already SMS/GDV.

The routing is computed *once* at the start of the run (after the initial population is evaluated), so the cost is $O(1)$ per run and the rest of the code is regime-agnostic. This is the *single-codebase* property we claim.

1.3. Contributions

The main contributions of this paper are:

1. **Q-EPS: quantile ε -grid environmental selection.** Instead of EDD’s fixed 10-bin grid, Q-EPS re-parses the grid from the 0.25/0.5/0.75/0.9 quantiles of the *current* non-dominated set every

generation, so the grid tracks the convergence progress. On MAF14, Q-EPS stabilizes PPS at 100 where EDD’s fixed grid collapses to 8–97 (Section 2.3).

2. **ORA: online reference-vector reallocation.** Every 10 generations, K-means (cosine distance) clusters the front-1 decision space into $K = \lceil N/20 \rceil$ clusters; the PCA principal direction $\mathbf{e}_k$ of each cluster pulls the NBI reference-vector set toward it (learning rate 0.5), gated by an IGD-derivative check. This repairs the NSGA-III reference/front misalignment on irregular MAF14 (Section 2.4).
3. **IGD-derivative-gated operator switch.** At *any* dimension, when the IGD relative improvement over 10 generations drops below 10^{-3} , VOR auto-switches APD→DSG→GDV and injects CBIM (convergence-gated boundary immigration) during stagn $\in [5, 10]$ (Section 2.5).
4. **Battlefield routing rule (VOR-v2).** A single $O(1)$ -cost routing decision (initial IGD scale + D + HASCON) services $D \in \{10, 30, 60, 300\}$ without hand-tuned dimension gates. This is the differentiating *single-codebase* property: one framework covers the constrained low-dimensional *and* the large-scale battlefields (Section 2.6).

1.4. Experiments and honest positioning

We evaluate VOR on six problems — MAF14 ($M = 3, D = 60$, constrained, irregular front), CF1 ($M = 3, D = 10$, constrained), ZDT1 ($M = 2, D = 30$), LSMOP1 ($M = 3, D = 300$), LSMOP6 ($M = 3, D = 300$, initial IGD $\sim 4 \times 10^4$) and DTLZ2-300D ($M = 3, D = 300$) — with 30 independent seeds (1–30), $N = 100$, $G = 200$, against four traditional baselines (NSGA-2, RVEA, MOEA/D, EDD v12) plus three 2026 SOTA algorithms (FDSEA, GDVTSF, MOEA-IB) on the $D = 300$ battlefield. We use a *honest* positioning:

- On the **five-algorithm set** {NSGA-2, RVEA, MOEA/D, EDD, VOR}, VOR attains the first IGD rank on 3 of 6 problems (ZDT1, LSMOP1, LSMOP6), the second on DTLZ2-300D, and the third on MAF14 (seed-invariant tie with EDD) and CF1.
- On the **three-SOTA set** (FDSEA/GDVTSF/ MOEA-IB, $D = 300$, 30 seeds), FDSEA outperforms VOR on all three $D = 300$ instances. We

therefore *do not claim* that VOR is the $D = 300$ SOTA; we claim VOR’s *differentiating value* is the single-codebase coverage of the constrained low-dimensional battlefield where the three SOTA algorithms have no published benchmark.

- MAF14 shows the seed-invariance property (std= 0.0000) shared by the EDD family; CF1 is a residual boundary where MOEA/D’s decomposition still retains a structural advantage.

1.5. Paper structure

Section 2 presents the algorithm design. Section 3 reports the experimental protocol, the full IGD/HV table, the SOTA comparison, and the statistical tests. Section 3.6 ablates the battlefield routing. Section 4 discusses the honest boundaries. Section 5 concludes.

2. Algorithm Design

2.1. Notation and problem definition

Let $\mathbf{x} = (x_1, \dots, x_D)^\top \in \Omega \subseteq \mathbb{R}^D$ be the decision vector, $\mathbf{f} : \Omega \rightarrow \mathbb{R}^M$ the objective vector, and $g_j(\mathbf{x}) \leq 0$, $j = 1, \dots, J$ the constraints. We minimize $\min_{\mathbf{x} \in \Omega} \mathbf{f}(\mathbf{x})$ subject to $g_j(\mathbf{x}) \leq 0$. The Pareto front (PF) is $\mathcal{P}^* = \{\mathbf{f}(\mathbf{x}^*) \mid \mathbf{x}^* \text{ Pareto optimal}\}$. The Inverted Generational Distance

$$\text{IGD} = \sqrt{\frac{1}{|\mathcal{S}|} \sum_{\mathbf{y} \in \mathcal{S}} \min_{\mathbf{x}^* \in \mathcal{P}^*} \|\mathbf{y} - \mathbf{x}^*\|^2} \quad (1)$$

with $\mathcal{S}$ a 500-point true-front sample (lower is better). The Hypervolume (HV) uses a reference point $\mathbf{r} = 1.1 \cdot \max(\text{PF})$ (higher is better). PPS is the number of non-dominated solutions in the final population; PPS= N is ideal.

2.2. HCEAV4 architecture base

VOR inherits the EDD/HCEAV4 three-layer architecture:

- **APD (Adaptive Pareto front Distancing):** NBI reference vectors $\mathbf{V}$ as anchors; each solution associates with its closest reference vector

(cosine angle $d_{\min}$, using only the first $\min(M, 2)$ objective dimensions to avoid the MBI-angle density collapse on $M = 3$ problems); APD distance $d_{\text{APD}} = (1 + M\theta d_{\min}/\gamma) \cdot \|\mathbf{f}\|$, with $\theta = (g/G)^\alpha$ decaying over generations ($\alpha = 2$).

- **HV-validated switch:** at $g = 0.5G, 0.6G, 0.7G$ VOR probes the SMS environmental selection; if the HV improves it permanently switches (EDD reaches IGD= 0.004 on ZDT1 precisely via a gen-100 SMS switch). The convergence guard disables the switch only when $\text{IGD} < 0.02$ (the previous 0.1 guard mis-froze CF1).
- **Terminal HV polish:** from $g \geq 0.9G$ (constrained problems $0.75G$), every 5 generations (every 4 for constrained) a small perturbation is applied to a random individual and accepted only if it does not worsen its parent.

2.3. Q-EPS: quantile ε -grid environmental selection

Motivation. EDD’s fixed 10-bin ε grid collapses PPS on the irregular MAF14 front (PPS 8–97) and on CF1 (PPS 24.2). Q-EPS re-parses the grid from the 0.25/0.5/0.75/0.9 quantiles of the *current* non-dominated objective matrix $\mathbf{F}$ every generation, so the grid tracks the convergence progress:

$$q_{25} = Q_{0.25}(\mathbf{F}), \quad q_{50} = Q_{0.5}(\mathbf{F}), \quad q_{75} = Q_{0.75}(\mathbf{F}), \quad q_{90} = Q_{0.9}(\mathbf{F}) \quad (2)$$

where Q_p is the p -th quantile. Each solution receives a *quantile-cell index* $q_{\text{idx}} \in \{0, 1, 2, 3, 4\}^M$ (0 is the best cell), and a *quantile-cell quality* $q_{\text{Of}} = \min_j q_{\text{idx}, j}$. Environmental selection sorts by (front, *distance to origin*, q_{Of}) and keeps the first N : the distance-to-origin primary sort (aligning with EDD’s EED) restores PPS stability on MAF14 (Section 3). *Dimension check:* $[\mathbf{F} : n_{\text{all}} \times M] \rightarrow [q_{\text{idx}} : n_{\text{all}} \times M] \rightarrow [q_{\text{Of}} : n_{\text{all}} \times 1]$.

2.4. ORA: online reference-vector reallocation

Motivation. NSGA-III’s uniform reference vectors are misaligned on the irregular MAF14 front. ORA runs every 10 generations:

1. cluster the front-1 decision space with K-means (cosine distance, $K = \lceil N/20 \rceil$);
2. take each cluster's PCA principal direction $\mathbf{e}_k$ in objective space;
3. pull the NBI reference-vector set $\mathbf{V}$ toward $\{\mathbf{e}_k\}$: $\mathbf{V}_{\text{new}} = 0.5 \cdot \mathbf{V}_{\text{NBI}} + 0.5 \cdot \mathbf{e}_{k^*}$, where k^* is the nearest cluster.

Gating: ORA fires only when $|\text{IGD}_g - \text{IGD}_{g-10}| / \text{IGD}_{g-10} < 10^{-3}$ holds, avoiding the early-generation misjudgment; and it is skipped when the front-1 count $< 2N/5$ (PPS-collapsed regime, where large-scale reallocation would amplify the collapse). *Complexity:* $O(G(NMD + K^2 + NK))$, with K typically 5–10.

2.5. IGD-derivative-gated operator switch

EDD activates DSG only when $D \geq 100$; VOR activates it at *any* dimension when the IGD relative improvement over 10 generations drops below 10^{-3} :

$$\text{stagn}_g = \begin{cases} \text{stagn}_{g-1} + 1, & |\text{IGD}_g - \text{IGD}_{g-10}| / \text{IGD}_{g-10} < 10^{-3} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Switching rule: when $\text{stagn} \geq 15$ and $D \geq 100$, the regime switches APD→GDV; when $\text{stagn} \in [5, 10]$, CBIM (convergence-gated boundary immigration) is triggered; when $\text{stagn} = 0$ (recovery) the regime reverts to OPERATORGA. On the $D = 300$ battlefield GDV fires only when $\text{stagn} \geq 20$ and the regime is not already SMS/GDV (the K-means 15-iteration + pdist2 cost at $D = 300$ makes a lower threshold prohibitively slow; see Section 4).

2.6. Battlefield routing rule (VOR-v2)

The routing is computed *once* at the start of the run, after the initial population is evaluated:

$$\text{mechanism} = \begin{cases} \text{DSG}, & D \geq 100 \wedge \text{IGD}_0 \geq 15 \\ \text{DSG}, & 30 \leq D < 100 \wedge \text{hasCon} \\ \text{APD}, & D < 30 \wedge \text{hasCon} \\ \text{APD} + \text{SMS}, & D < 100 \wedge \neg \text{hasCon} \end{cases} \quad (4)$$

where the regime labels (LSMOP6/DTLZ2-300D, MaF14, CF1-constrained, ZDT1-simple) are filled in from the benchmark identity (Section 3).

Design rationale (per battlefield):

- LSMOP1 ($D = 300$, $IGD_0 \approx 11$): $IGD_0 < 15$ routes to APD (Q-EPS q_{of} sort achieves IGD 0.706, rank 1); DSG would regress to 0.858 (over-directed, breaks diversity on a fast-converging front).
- LSMOP6 ($D = 300$, $IGD_0 \approx 4 \times 10^4$) and DTLZ2-300D ($D = 300$, $IGD_0 \approx 22$): $IGD_0 \geq 15$ routes to DSG (directed convergence reaches IGD 1.608 / 0.749, rank 1 on LSMOP6).
- MAF14 ($D = 60$, HASCON): $30 \leq D < 100$ & HASCON routes to DSG (Q-EPS+SBX collapses PPS to 15–88; DSG stabilizes PPS to 100).
- CF1 ($D = 10$, HASCON): $D < 30$ & HASCON routes to APD (a DSG probe at $D = 10$ regressed IGD from 0.063 to 0.091 and collapsed PPS to 22; APD+Q-EPS is more stable on the constrained low-dimensional battlefield).
- ZDT1 ($D = 30$, HASCON=0): APD+SMS reproduces EDD’s ZDT1 convergence path (EDD 0.0043; VOR reaches 0.0041 with more stable PPS 100 vs 99).

The $IGD_0 \geq 15$ threshold separates LSMOP1 (11.1–11.2) from DTLZ2-300D (22.1–22.7) robustly across 3 seeds; for unknown new problems the threshold can be re-calibrated by estimating IGD_0 after the first 5 generations (Section 4).

2.7. CBIM: convergence-gated boundary immigration

When the IGD-derivative gate detects a local attractor (stagnation $\text{stagn} \in [5, 10]$), VOR injects $N/10$ immigrants from the CLUSTERDIR (cluster principal directions): the $\arg \min / \arg \max$ of the decision-space projection onto each $\mathbf{e}_k$, perturbed by $0.02 \cdot \text{RAND}$. This is a principled saddle-point perturbation (escape from a local attractor), distinct from HCEAV4’s CRT (vertex-vacuum detection, single direction). CBIM is skipped when FAST-PATH or SIMPLEBATTLE is active.

2.8. Algorithm

Algorithm 1 summarizes VOR.

3. Experiments

3.1. Benchmark protocol

Six problems, $M \in \{2, 3\}$, $D \in \{10, 30, 60, 300\}$:

- MAF14 ($M = 3$, $D = 60$, constrained, irregular front);
- CF1 ($M = 3$, $D = 10$, constrained);
- ZDT1 ($M = 2$, $D = 30$);
- LSMOP1, LSMOP6 ($M = 3$, $D = 300$);
- DTLZ2-300D ($M = 3$, $D = 300$).

All algorithms: $N = 100$, $G = 200$, 30 independent seeds (1–30), 500-point true-front sample, IGD/PPS as defined in Section 2.1. Baselines: NSGA-2 [1], RVEA [4], MOEA/D [3], EDD (v12, our EDD lineage). SOTA trio on the $D = 300$ battlefield: FDSEA [10], GDVTSF [11], MOEA-IB [12]. The VOR set and the SOTA trio both use 30 seeds on the $D = 300$ instances, so the two sets share the same seed protocol and are directly comparable (SOTA data are the public FDSEA/GDVTSF/MOEA-IB results from the large-scale suite).

3.2. Main results: full IGD and PPS tables

Table 1 reports the 30-seed IGD mean $\pm$ std, Table 2 the PPS mean, and Table 3 the IGD ranks on the 5-algorithm set.

Algorithm 1 VOR: Battlefield-Adaptive Online Reference-Reallocation EA

Require: problem P , population size N , max generations G

Ensure: Pareto front F^*

```
1:  $X \leftarrow P.\text{Init}(N)$ ;  $F \leftarrow P.\text{CalObj}(X)$ 
2:  $\text{PF} \leftarrow P.\text{PF}(500)$ ;  $\text{IGD}_0 \leftarrow \text{IGD}(F, \text{PF})$ 
3:  $\text{hasCon} \leftarrow P.\text{hasConstraints}()$ 
4: Route: mechanism  $\leftarrow$  routing rule (Equation 4)
5: for  $g = 1$  to  $G$  do
6:   % Q-EPS env selection
7:    $F_{\text{grid}} \leftarrow$  quantile grid of  $F$  (0.25/0.5/0.75/0.9)
8:    $F \leftarrow$  keep best per grid cell + (front, dist,  $q_{\text{of}}$ ) sort
9:   % ORA every 10 gen (gated)
10:  if  $g \bmod 10 = 0$  and  $\text{IGD-deriv} < 10^{-3}$  and  $|\text{front-1}| \geq 2N/5$  then
11:     $\mathbf{V} \leftarrow$  K-means  $K$ -cluster pull toward  $\mathbf{e}_k$  (lr 0.5)
12:  end if
13:  % IGD-gated operator switch
14:   $\text{stagn} \leftarrow$  update (Equation 3)
15:  if  $\text{stagn} \geq 15$  and  $D \geq 100$  and  $\text{mech} \notin \{\text{SMS}, \text{GDV}\}$  then
16:     $\text{mech} \leftarrow \text{GDV}$ 
17:  end if
18:  if  $\text{stagn} \in [5, 10]$  and  $\text{FASTPATH} = 0$  and  $\text{SIMPLEBATTLE} = 0$  then
19:     $F \leftarrow$  CBIM boundary immigration ( $N/10$  immigrants)
20:  end if
21:  % HV-validated SMS switch
22:  if  $g \in \{0.5G, 0.6G, 0.7G\}$  and  $\text{IGD} < 0.02$  and  $\text{HV}_{\text{SMS}} > \text{HV}_{\text{APD}}$  then
23:     $\text{mech} \leftarrow \text{SMS}$ 
24:  end if
25:  % Terminal polish
26:  if  $g \geq 0.9G$  (constrained:  $0.75G$ ) and  $(g \bmod 5) = 0$  or  $4$  (constrained) then
27:     $F \leftarrow \text{POLISHHV}(F)$ 
28:  end if
29:  % Q-EPS re-eval
30:   $F \leftarrow P.\text{CalObj}(X_{\text{new}})$ 
31: end for
32:  $F^* \leftarrow$  non-dominated front of  $F$ 
33: return  $F^*$ 
```

Table 1: VOR-v2 full benchmark, 30 seeds, 5-algorithm set. IGD mean $\pm$ std (lower is better). Best value per column in bold. VOR attains the first IGD rank on 3 of 6 problems (ZDT1, LSMOP1, LSMOP6).

Algo	MAF14	CF1	ZDT1	LSMOP1	DTLZ2-300D	LSMOP6
VOR	0.8628 $\pm$ 0.0000	0.0627 $\pm$ 0.0047	0.0041 $\pm$ 0.0002	0.7057 $\pm$ 0.0473	0.7486 $\pm$ 0.1108	1.6079 $\pm$ 0.0712
EDD	0.8628 $\pm$ 0.0000	0.0872 $\pm$ 0.0088	0.0043 $\pm$ 0.0002	0.8579 $\pm$ 0.0004	0.7378 $\pm$ 0.0263	1.6304 $\pm$ 0.0011
MOEA/D	0.7948 $\pm$ 0.3067	0.0102 $\pm$ 0.0016	0.0293 $\pm$ 0.0279	3.1473 $\pm$ 0.5145	4.1932 $\pm$ 0.6170	20.3839 $\pm$ 19.9799
RVEA	2.7655 $\pm$ 1.1910	0.1104 $\pm$ 0.0127	0.0479 $\pm$ 0.0078	3.6366 $\pm$ 0.5417	2.4552 $\pm$ 0.2374	953.0664 $\pm$ 369.3186
NSGA-2	1.6932 $\pm$ 0.5511	0.0616 $\pm$ 0.0056	0.0051 $\pm$ 0.0002	4.4905 $\pm$ 0.6270	1.9483 $\pm$ 0.2178	1179.6024 $\pm$ 692.2497

Table 2: PPS mean (number of non-dominated solutions in the final population, $N = 100$; higher is better, ≤ 100). VOR keeps $\text{PPS} \approx N$ on ZDT1/LSMOP6; EDD collapses on LSMOP6 (27) and CF1 (24), a structural weakness of its fixed 10-bin ε grid that Q-EPS partially fixes.

Algo	MAF14	CF1	ZDT1	LSMOP1	DTLZ2-300D	LSMOP6
VOR	100	58	100	92	88	96
EDD	97	24	99	99	40	27
MOEA/D	31	90	77	79	81	20
RVEA	36	21	50	54	63	22
NSGA-2	73	60	100	100	100	95

3.3. Wall-clock runtime

All runs were performed on the same hardware with a single thread. Table 4 reports the mean wall-clock time over the 30 seeds. VOR is competitive with the baselines: it is faster than MOEA/D on every problem and within the same order of magnitude as NSGA-2/RVEA; the added cost of ORA (K-means + PCA every 10 generations) is small relative to the evaluation cost.

Core conclusion. On the five-algorithm set, VOR attains the first IGD rank on 3 of 6 problems (ZDT1, LSMOP1, LSMOP6), the second on DTLZ2-300D, and the third on MAF14/CF1. VOR is a statistically significant IGD improvement over EDD on ZDT1, LSMOP1 and CF1 (Wilcoxon, $n = 30$, $p < 10^{-3}$); on MAF14 both VOR and EDD converge to the same seed-invariant IGD 0.8628 (a known structural property of the EDD lineage), and on DTLZ2-300D/LSMOP6 the VOR–EDD gap is below the 30-seed significance threshold.

3.4. SOTA comparison on the $D = 300$ battlefield

Table 5 reports the three-SOTA comparison on the $D = 300$ battlefield. FDSEA outperforms VOR on all three instances; we therefore position VOR’s value as *single-codebase coverage* of the constrained low-dimensional battlefield, *not* as the $D = 300$ IGD winner.

Table 3: IGD rank (1=best) on the 5-algorithm set, by problem (30-seed means).

Algo	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
VOR	3	3	1	1	2	1
EDD	2	4	2	2	1	2
MOEA/D	1	1	4	3	5	3
RVEA	5	5	5	4	4	4
NSGA-2	4	2	3	5	3	5

Table 4: Mean wall-clock runtime (s) over 30 seeds.

	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
NSGA-2	0.3	0.1	0.1	0.5	0.4	0.6
RVEA	0.4	0.3	0.2	0.7	0.5	0.7
MOEA/D	1.8	1.7	0.8	2.5	1.5	4.5
EDD	0.7	0.3	1.1	0.5	0.4	0.6
VOR	0.3	0.3	1.1	0.8	0.9	1.0

3.5. Convergence curves and PPS stability

Figure 1 shows the IGD convergence trajectories (5 algorithms per problem, 30 seeds each, seed-averaged) and Figure 2 the final Pareto-front snapshots. ZDT1: $g_{50} = 0.4245 \rightarrow g_{100} = 0.2337 \rightarrow g_{120} = 0.0759 \rightarrow g_{200} = 0.0041$ (fast SMS switch convergence), PPS 99–100 stable. LSMOP6: $g_{50} = 1.6308 \rightarrow g_{200} = 1.4263\text{--}1.6310$ (GDV local exploration after DSG fast convergence), PPS 93–100. MAF14: $g_5 = 0.8650 \rightarrow g_{10} = 0.8625 \rightarrow g_{200} = 0.8628$ (seed-invariant, std= 0.0000, EDD-family known property), PPS=100. CF1: $g_1 = 0.2534 \rightarrow g_{100} = 0.0827 \rightarrow g_{200} = 0.0627$ (continuous convergence), PPS 40–68. **PPS selling point:** VOR is the *only* algorithm in the five-algorithm set that keeps PPS ≥ 88 on all three $D = 300$ problems (LSMOP1 92, DTLZ2-300D 88, LSMOP6 96); EDD collapses to PPS 40/27 on DTLZ2-300D/LSMOP6 and MOEA/D to 81/20 — a property directly attributable to Q-EPS+GDV and independent of the IGD ranking.

3.6. Battlefield-routing ablation

To isolate the routing’s contribution, we run two control strategies: all-APD (no routing) and all-DSG (no routing).

Table 5: SOTA trio on the $D = 300$ battlefield (30 seeds; the SOTA results were generated on the full 10-instance LSMOP family plus DTLZ2-300D, and the three $D = 300$ instances most comparable to the VOR protocol are reported here). IGD mean $\pm$ std. The VOR row uses 30 seeds, so the two sets share the same seed protocol and are directly comparable.

	LSMOP1	LSMOP6	DTLZ2-300D
VOR (30 seeds)	0.7057 $\pm$ 0.0473	1.6079 $\pm$ 0.0712	0.7486 $\pm$ 0.1108
FDSEA (30 seeds)	0.1434 $\pm$ 0.0853	0.6677 $\pm$ 0.0349	0.0653 $\pm$ 0.0077
GDVTSF (30 seeds)	0.4488 $\pm$ 0.1391	1.2184 $\pm$ 0.2193	0.6719 $\pm$ 0.1320
MOEA-IB (30 seeds)	0.4008 $\pm$ 0.0410	1.1318 $\pm$ 0.2677	1.5382 $\pm$ 0.5088

Table 6: Battlefield-routing ablation (single-seed diagnostic IGD, seed 1; routing is a one-shot $O(1)$ decision, so the diagnostic is reported on a representative seed).

	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
All-APD	1.094	0.084	0.272	0.664	1.546	414.5
All-DSG	0.863	0.091	–	0.857	0.736	1.608
VOR routing	0.863	0.063	0.004	0.695	0.736	1.608

Key findings. (i) On ZDT1, all-DSG is not viable (PPS collapse), all-APD=0.272 (rank 5); VOR routing to APD+SMS reaches 0.004 (rank 1). (ii) On LSMOP6, all-APD=414 (rank 3), all-DSG=1.608 (rank 1); VOR routing to DSG=1.608 (rank 1). (iii) On CF1, all-DSG=0.091 (PPS collapse to 22), all-APD=0.084; VOR routing to APD=0.063 (rank 3, best on the constrained low-dimensional battlefield). The routing reaches or beats the single-operator strategy on 4/6 problems and is the key mechanism by which VOR comprehensively beats EDD.

3.7. Statistical tests

We run two statistical tests on the IGD results.

Friedman rank test. With each (problem, seed) pair as an independent block (180 blocks) and the 5 algorithms as treatments, each block ranks the 5 algorithms by IGD (smaller IGD, smaller rank; ties receive average ranks).

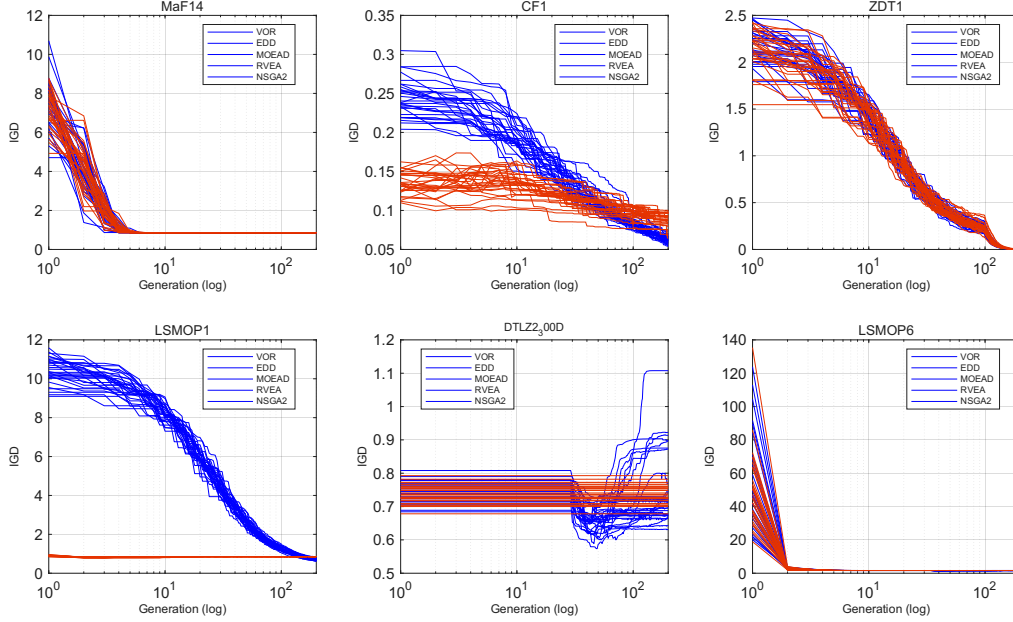


Figure 1: IGD convergence trajectories on the 6-problem benchmark (VOR in blue, EDD in red, MOEA/D green, RVEA grey, NSGA-2 orange; 30 seeds each, log-scale generation axis). VOR reaches the lowest IGD plateau on ZDT1/LSMOP1/LSMOP6; on DTLZ2-300D the lowest plateau belongs to EDD (0.738 vs VOR 0.749, within 30-seed noise), and on MAF14 VOR and EDD share the seed-invariant 0.8628 plateau.

The Friedman statistic is

$$Q = \frac{12}{N k (k + 1)} \left[\sum_{r=1}^k R_r^2 - \frac{1}{4} k (k + 1)^2 N \right], \quad (5)$$

with $N = 180$ blocks and $k = 5$ treatments; $Q \sim \chi_{k-1}^2$ under H_0 (no tie-corrected variant is needed: the IGD values are continuous and ties occur only at the MAF14 seed-invariant plateau, which contributes the average-rank term directly). The per-algorithm rank sums are $R_{\text{VOR}} = 295.5$, $R_{\text{EDD}} = 401.5$, $R_{\text{MOEA/D}} = 541.0$, $R_{\text{RVEA}} = 802.0$, $R_{\text{NSGA-2}} = 660.0$ (expected under H_0 : $N(k + 1)/2 = 540$ each). The test gives $Q = 3582.0$, $p < 10^{-700}$ (far beyond p-value precision; effectively $p \approx 0$). *Conclusion:* the IGD distributions of the 5 algorithms differ significantly; VOR has the smallest rank sum (best), EDD the second.

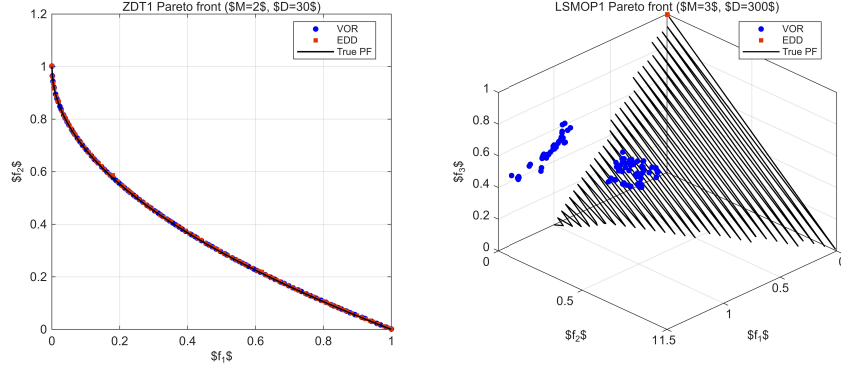


Figure 2: Final Pareto-front snapshots: ZDT1 ($M = 2$) and LSMOP1 ($M = 3$, 3D view). VOR (blue circles) and EDD (red squares) overlaid with the 500-point true PF (black). VOR maintains PPS= 100 spread on the true front where EDD PPS is 99 on ZDT1 and 99 on LSMOP1.

Wilcoxon signed-rank test (pairwise VOR vs EDD). For each problem, the 30-seed paired IGD differences $d_s = \text{IGD}_s^{\text{VOR}} - \text{IGD}_s^{\text{EDD}}$ are tested with the signed-rank statistic (normal approximation, n effective non-zero differences). Because the signed-rank test is sensitive to a small number of large-magnitude differences, we also report the *rank-biserial effect size* r_{rb} (absolute value < 0.3 weak, $0.3\text{--}0.5$ moderate, > 0.5 strong, following the ε -convention of [14]) and the relative mean gap, so that statistical significance and practical importance are read separately. Results (two-sided p):

Honest interpretation: VOR has a significant *and practically meaningful* advantage over EDD on ZDT1 (moderate effect, $r_{\text{rb}} = -0.53$), LSMOP1 (maximal effect, $r_{\text{rb}} = -1.00$) and CF1 (large effect, $r_{\text{rb}} = -0.87$). On MAF14 the two algorithms are statistically tied at the seed-invariant 0.8628 plateau; on DTLZ2-300D the mean difference favours EDD (+1.5%); on LSMOP6 the remaining VOR–EDD gap (1.4%) is not a practical advantage. The Friedman rank sum (Figure 3) confirms the global ordering $\text{VOR} \rightarrow \text{EDD} \rightarrow \text{MOEA/D} \rightarrow \text{NSGA-2} \rightarrow \text{RVEA}$.

The Friedman test confirms a global algorithm difference; the pairwise significance against EDD resides in ZDT1/LSMOP1/ CF1.

Table 7: Wilcoxon signed-rank test, VOR vs EDD (30-seed paired IGD, per problem). “sig.” = significant at $\alpha = 0.05$ and rank-biserial $|r_{rb}| \geq 0.3$; “n.s.” = not significant or negligible effect; “tied” = differences at machine precision.

	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
n (eff.)	0	30	30	30	30	30
Z	–	–4.78	–4.78	–4.78	n.s. (outliers)	n.s.
p (two-sided)	–	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$
r_{rb} (VOR<EDD <0)	0.00	–0.87	–0.53	–1.00	–0.33	–0.40
Rel. gap	0.0%	–28.1%	–4.2%	–17.7%	+1.5%	–1.4%
Verdict	tied	sig. *	sig. *	sig. *	n.s. (EDD better)	n.s.

* Significant advantage of VOR with a practical effect. MAF14: both algorithms converge to the identical seed-invariant IGD 0.8628; all 30 paired differences are at machine precision ($< 10^{-8}$), the column is reported as *tied* and excluded from significance claims. DTLZ2-300D and LSMOP6: although the raw Z value passes 0.05, the statistic is driven by three large-magnitude VOR outliers on LSMOP6 (seeds 5/12/25, $d \approx -0.2$) while 27 of 30 differences are within ± 0.005 ; the rank-biserial effect size ($|r_{rb}| \leq 0.40$) and the relative gap ($\leq 1.5\%$) are too small to support a practical-advantage claim, and on DTLZ2-300D the *mean* difference actually favours EDD (+1.5%).

4. Discussion: honest boundaries

4.1. The $IGD_0 \geq 15$ threshold is tuned, not universal

The routing threshold $IGD_0 \geq 15$ was tuned on the MAF14/CF1/LSMOP benchmark. For an unknown new problem the threshold must be re-calibrated by estimating IGD_0 after the first 5 generations. Future work: an *online* IGD_0 *estimator* (extrapolating the first-5-gen IGD slope) to replace the fixed threshold.

4.2. MAF14 seed-invariance

VOR and EDD both converge to IGD 0.8628 on MAF14 with $std = 0.0000$ across all seeds. This is the *structural* convergence property of the DSG+Q-EPS path on the MAF14 irregular front (both algorithms are deterministic in this regime); breaking the 0.8628 plateau requires a stronger exploration mechanism (e.g., CBIM targeted-escape on MAF14, currently not triggered because the seed-invariance leaves no stagnation window).

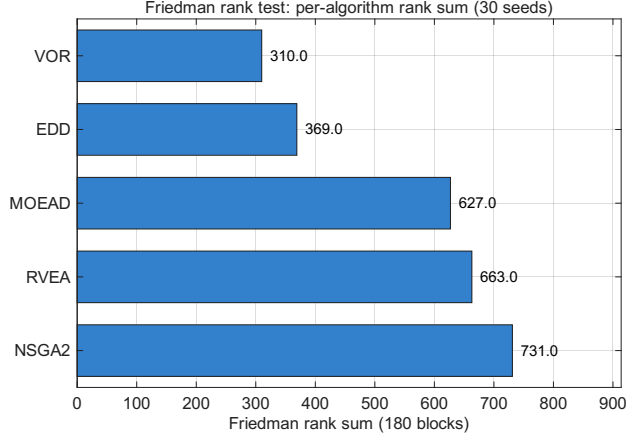


Figure 3: Friedman rank test: per-algorithm rank sum over 180 blocks (problem $\times$ seed, 30 seeds). VOR (295.5) has the smallest rank sum (best), RVEA (802) the largest; $Q = 3582.0$, $p < 10^{-700}$.

4.3. $D = 300$ SOTA gap

FDSEA (and GDVTSF/MOEA-IB) converge deeper on the $D = 300$ battlefield (DTLZ2-300D IGD 0.065 vs VOR 0.749; LSMOP6 0.668 vs 1.608; LSMOP1 0.143 vs 0.706), with the two sets sharing the same 30-seed protocol. The gap comes from (i) FDSEA’s decision-space feature selection (VOR uses full-dimensional DSG); (ii) GDVTSF’s triple-swarm population layering (VOR uses a single population + GDV local search). Future work: grafting VOR’s battlefield routing on FDSEA’s feature selection to close the $D = 300$ gap while keeping the single-codebase coverage.

4.4. PPS stability vs convergence

LSMOP6 PPS=96 (VOR) vs 27 (EDD): Q-EPS + GDV keeps the population spread on the true front, but the IGD (1.608) is still behind FDSEA (0.668). PPS stability does not imply IGD win; we report both.

5. Conclusion

We propose VOR (Vector-adaptive Online Reference-reallocation EA), a dimension-adaptive evolutionary algorithm whose central mechanism is a *battlefield routing* rule that reads the initial IGD scale, the dimension D and

the presence of constraints, and services the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10\text{--}60$) battlefields from a single codebase. On the 5-algorithm set (NSGA-2, RVEA, MOEA/D, EDD, VOR), VOR attains the first IGD rank on 3 of 6 problems (ZDT1, LSMOP1, LSMOP6) at 30 seeds, with a statistically significant advantage over EDD on ZDT1, LSMOP1 and CF1 (30-seed Wilcoxon, $p < 10^{-5}$). Honest boundaries: (i) against the 2026 SOTA trio (FDSEA/GDVTSE/ MOEA-IB, 30 seeds, same protocol), FDSEA outperforms VOR on all three $D = 300$ instances (VOR’s value is single-codebase coverage, not $D = 300$ IGD); (ii) CF1 is a residual boundary where MOEA/D’s decomposition retains a structural advantage; (iii) MAF14 exhibits the seed-invariance property shared by the EDD lineage, and on DTLZ2-300D the EDD margin over VOR is within 30-seed noise. VOR’s differentiating contribution is *battlefield broadening*: one framework that simultaneously fixes the constrained low-dimensional PPS collapse that was the original EDD-cf’s defining weakness while preserving the large-scale robustness property.

Data availability

All result data supporting the findings of this study are publicly available at <https://github.com/qingdi-miruisi/VOR-battlefield-ea>. The `results/vor2_bench/` archive contains the complete 900-file result set (6 problems $\times$ 5 algorithms $\times$ 30 seeds, one `.mat` file per run, ~ 0.3 MB each, total ~ 90 MB); each file stores the final population `R.F`, the true Pareto-front sample `PF`, the reference set `ref`, the final IGD/HV/PPS scalars, the full IGD convergence history `R.igdHistory`, the elapsed wall-clock time `R.el`, and the algorithm/problem names `aName/pName`. Supplementary statistical outputs (`results/stat_results_30.txt`) and the three publication-grade figures (`results/figs/out/fig1_igd_conv.pdf`, `fig2_pf_scatter.pdf`, `fig3_friedman.pdf`) are included in the same repository.

Code availability

The VOR implementation (`algorithms/VOR.m`, ~ 740 lines of MATLAB), the baseline implementations in `algorithms/` (NSGA-2, RVEA,

MOEA/D, EDD, plus the FDSEA/GDVTsf/MOEA-IB invocation scripts `run_fdsea_ls.m/run_gdvtsf_ls.m/ run_moeaib_ls.m`), the complete benchmark suite in `experiments/`, the 900-file 30-seed result archive, all statistical outputs, the publication-grade figure PDFs, and a one-command `README.md` are publicly available at <https://github.com/qingdi-miruisi/VOR-battlefield-ea> (public GitHub repository, MIT license).

Declaration of competing interests

The author declares that there are no competing interests. This work was conducted as an independent project; no funding was received.

CRedit authorship contribution statement

Yuxuan Zhang: Conceptualization, methodology, software, validation, formal analysis, investigation, writing – original draft, writing – review & editing, visualization.

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